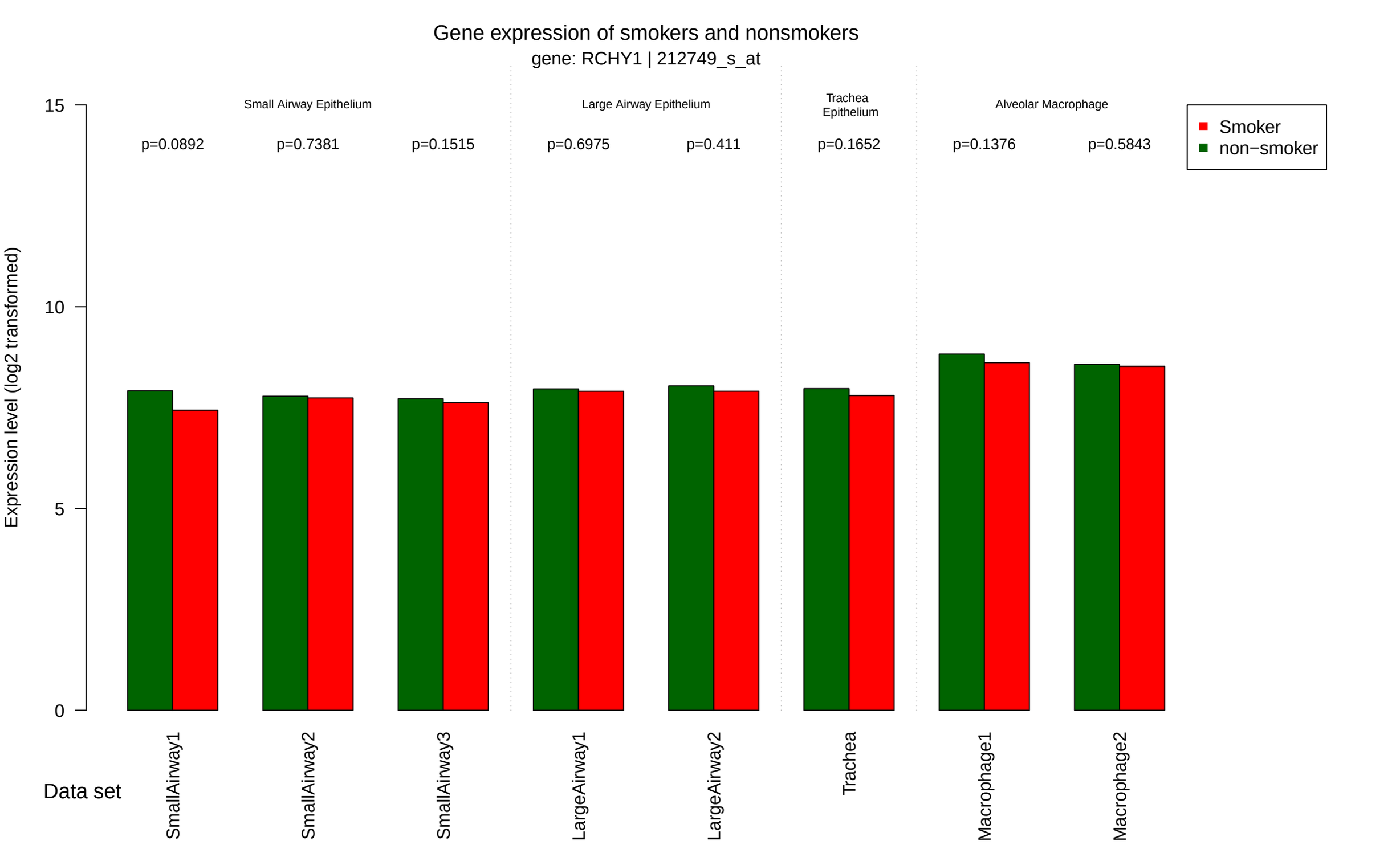
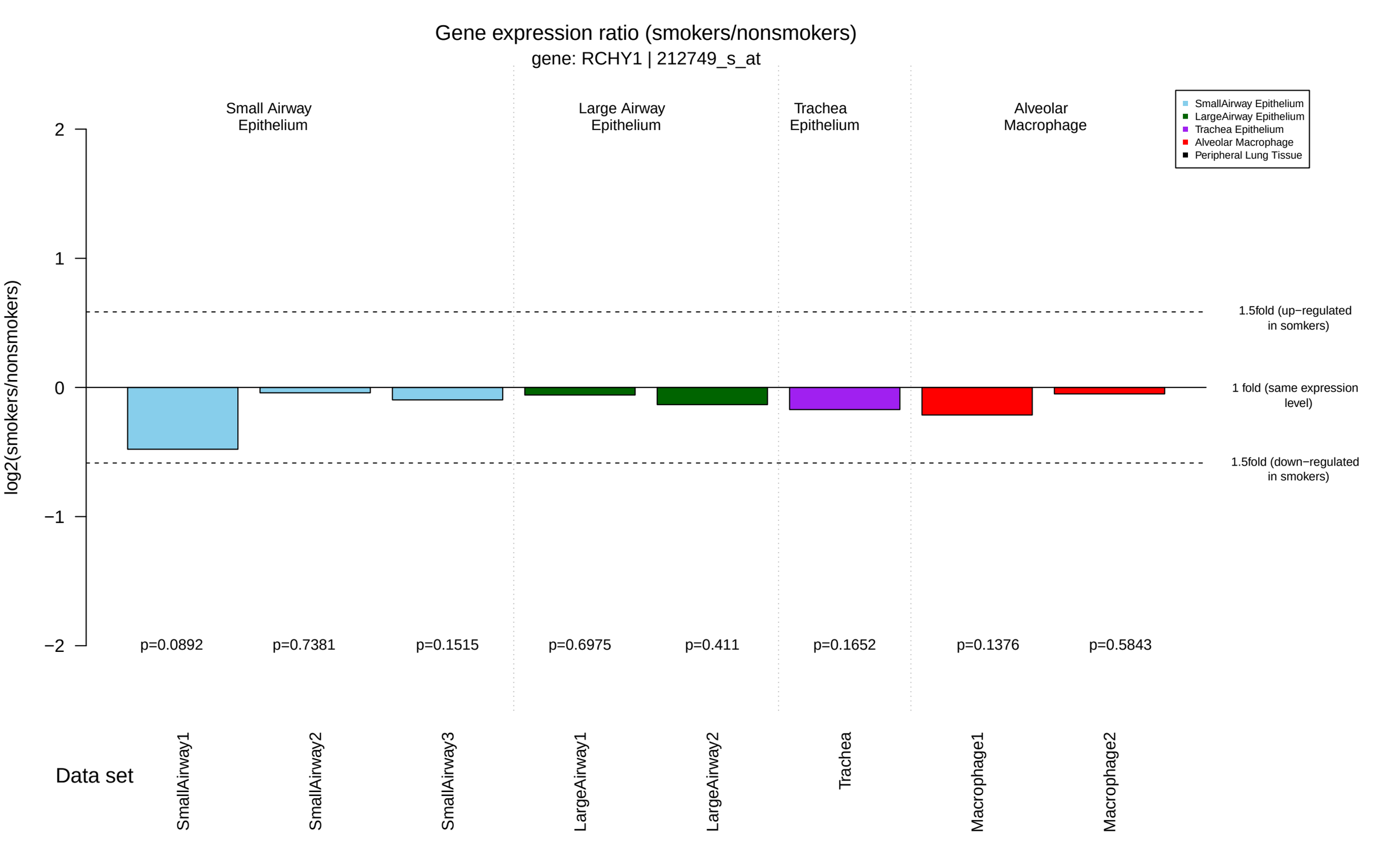
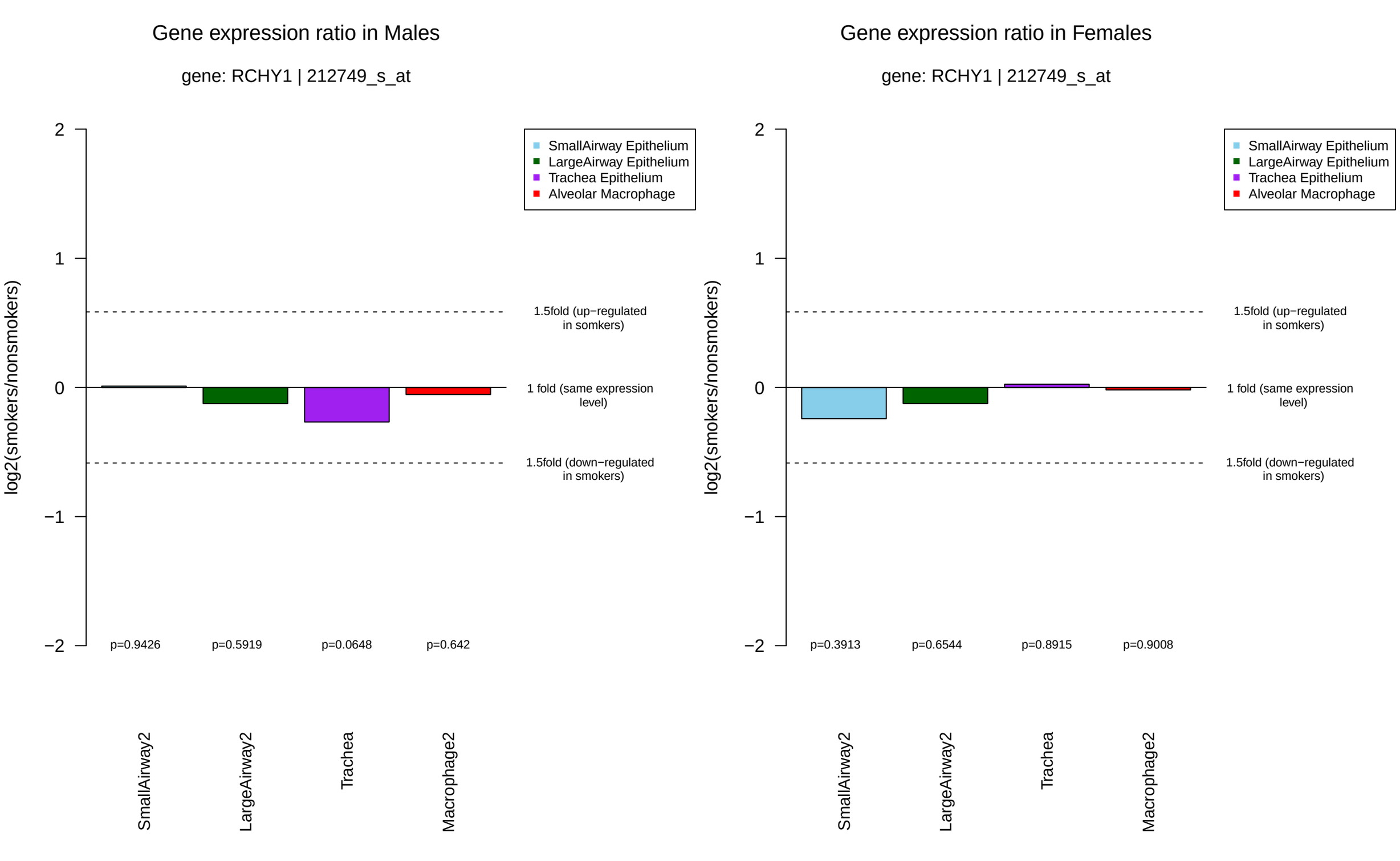
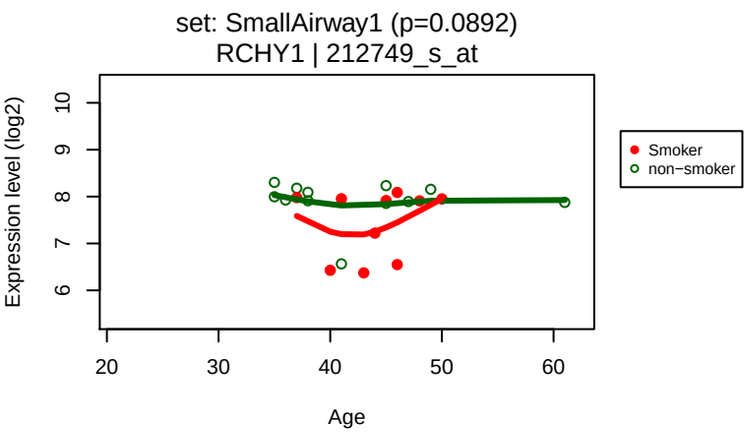
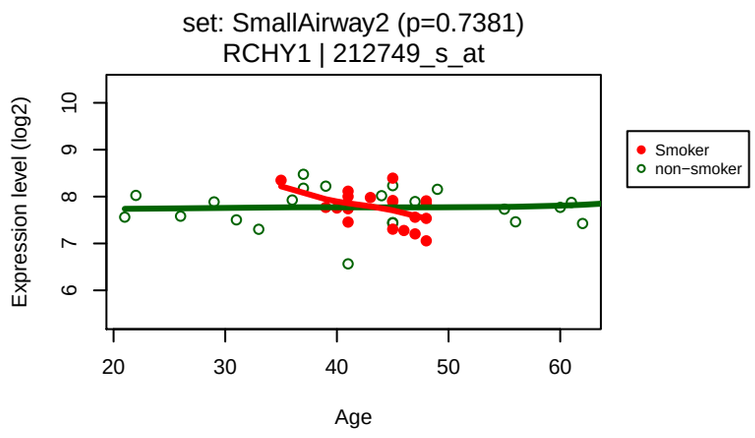
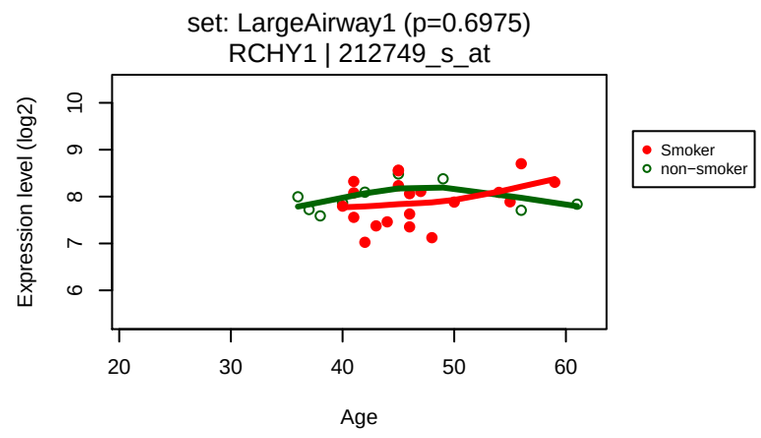

Gene expression of smokers and nonsmokers
gene: RCHY1 | 212749_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0892
p=0.7381
p=0.1515
p=0.6975
p=0.411
p=0.1652
p=0.1376
p=0.5843
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

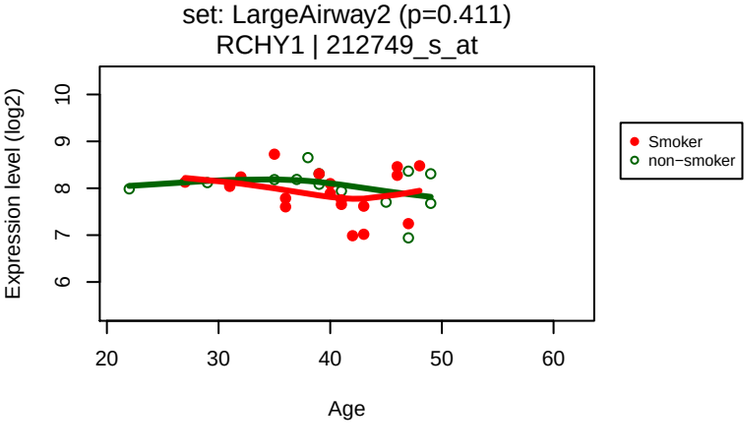

Gene expression ratio (smokers/nonsmokers)
gene: RCHY1 | 212749_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0892
p=0.7381
p=0.1515
p=0.6975
p=0.411
p=0.1652
p=0.1376
p=0.5843
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

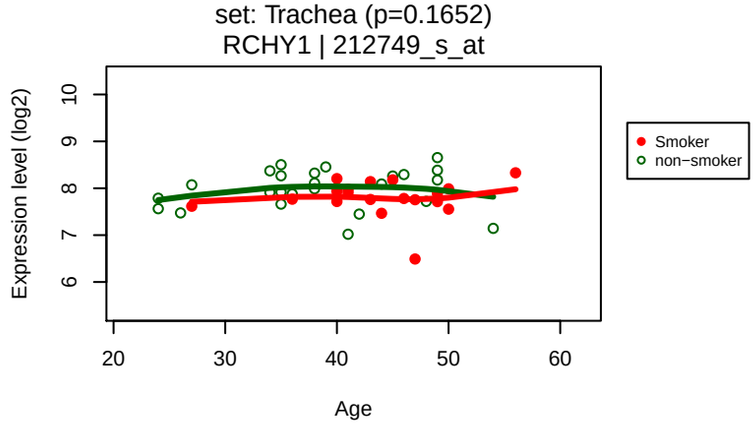

Gene expression ratio in Males
gene: RCHY1 | 212749_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9426
p=0.5919
p=0.0648
p=0.642
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RCHY1 | 212749_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3913
p=0.6544
p=0.8915
p=0.9008
SmallAirway2
LargeAirway2
Trachea
Macrophage2

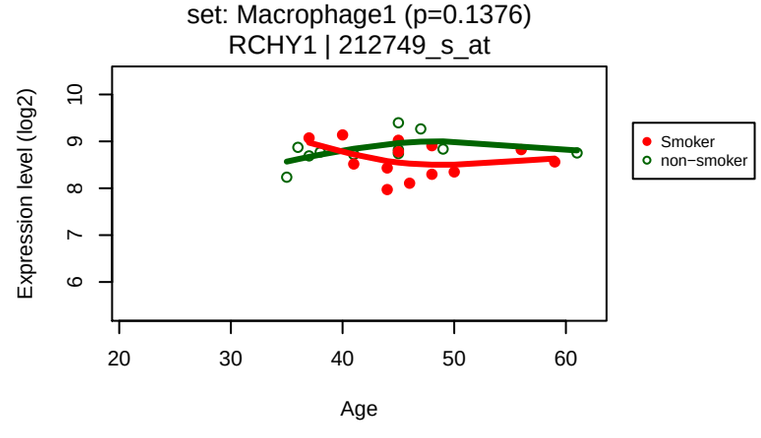

set: SmallAirway1 (p=0.0892)
RCHY1 | 212749_s_at
Expression level (log2)
Age
Smoker
non−smoker

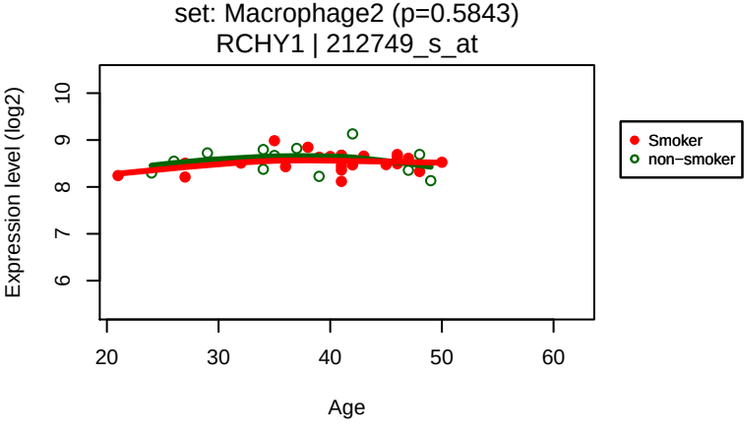

set: SmallAirway2 (p=0.7381)
RCHY1 | 212749_s_at
Expression level (log2)
Age
Smoker
non−smoker

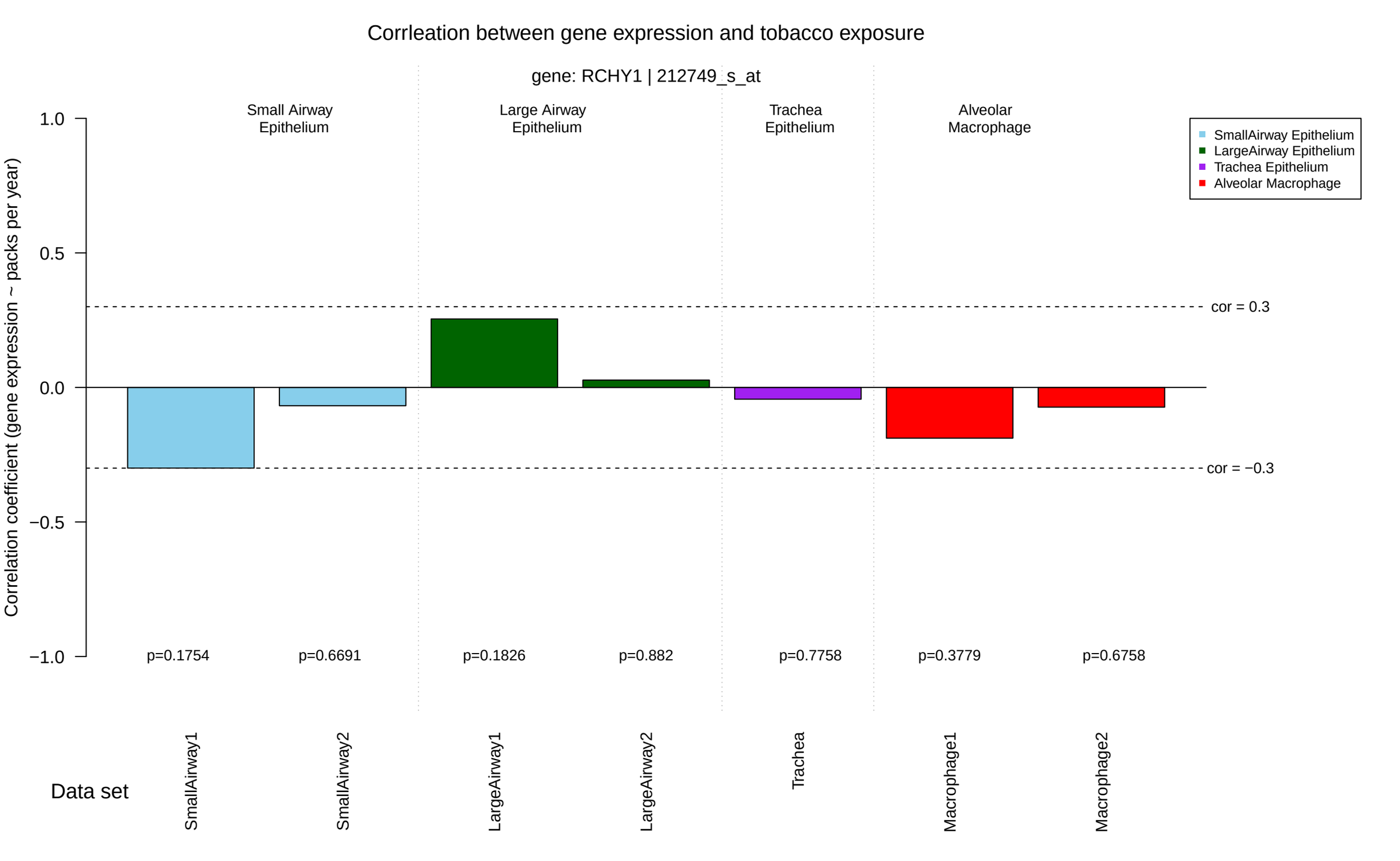

set: LargeAirway1 (p=0.6975)
RCHY1 | 212749_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.411)
RCHY1 | 212749_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1652)
RCHY1 | 212749_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1376)
RCHY1 | 212749_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5843)
RCHY1 | 212749_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RCHY1 | 212749_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1754
p=0.6691
p=0.1826
p=0.882
p=0.7758
p=0.3779
p=0.6758
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2